

Security Incident Report

Generated: 2026-07-10 15:26:58

Generated by AutoForensicAgent · Anomaly Detection & Digital Forensics Pipeline

UTC

Detected: 2026-07-10T15:26:29.380000

HIGH SEVERITY — Application Layer Protocol (T1071)

Anomaly Score: 0.628

EXECUTIVE SUMMARY

Device **IOT-SENSOR-031** (Temp Sensor, Xiaomi) at **192.168.1.63** was flagged for behavior consistent with **Application Layer Protocol** (MITRE ATT&CK T1071, tactic: Command and Control). The anomaly detection engine assigned a confidence score of **0.628**, and traffic evidence was automatically captured for forensic review under chain-of-custody ID **COC-ALERT-20260710-5A7994**.

DEVICE INFORMATION

DEVICE ID	IOT-SENSOR-031
DEVICE TYPE	Temp Sensor
VENDOR / FINGERPRINT	Xiaomi
IP ADDRESS	192.168.1.63
MAC ADDRESS	78:11:DC:09:E3:5A
FIRST SEEN ON NETWORK	2026-06-01T09:45:00Z

DETECTION FEATURES

PACKET RATE	1.8
BYTE RATE	598.1
UNIQUE DST IPS	1
UNIQUE DST PORTS	3
PORT ENTROPY	1.4591479170272446
SYN RATIO	0.0
UDP RATIO	0.0
AVG PACKET SIZE	332.27777777777777
FAILED AUTH COUNT	0
NEW DST RATIO	1.0

INCIDENT TIMELINE

- 2026-07-10T15:30:29.151524+00:00
First packet observed from IOT-SENSOR-031
- 2026-07-10T15:30:38.595968+00:00
Traffic window closed (18 packets analyzed)
- 2026-07-10T15:26:29.380412+00:00
Anomaly detector flagged window as 'c2_communication'

MITRE ATT&CK MAPPING

T1071 – Application Layer Protocol

Command and Control

TACTIC: Command and Control

Adversaries may communicate using application layer protocols to avoid detection by blending in with existing traffic, often indicating the device has already been compromised and is receiving instructions from a remote server.

Recommended Actions

- Immediately isolate the affected device from the network
- Block identified C2 IP/domain at the firewall
- Perform full firmware reset and forensic imaging before restoring device

EVIDENCE & CHAIN OF CUSTODY

PCAP FILE: evidence/ALERT-20260710-5A7994_capture.pcap
SHA-256: 8013cea2958d9b1da2ccdda5aa8f2aafffde7870e088126b862df91507a05984
CHAIN OF CUSTODY ID: COC-ALERT-20260710-5A7994
CAPTURED BY: NetGuard AutoForensicAgent v1.0

This report was generated automatically by the IoT Anomaly Detection & Digital Forensics module. It is intended to support, not replace, human security analyst review. Evidence integrity should be verified by re-computing the SHA-256 hash of the referenced pcap file before use in any formal investigation.

REPORT INTEGRITY HASH (SHA-256): 9911e6b0f5e8238566e8614fef0532cf2fe8b4668889653acf3b7069c29846bf